March 12, 2026

Recap of sampling / simulations so far

- sampling / simulations using built-in functions
  - sample( )
  - r- ( )
    p- ( )
    d- ( )
    q- ( )

    for a family of distributions

- What if there is no built-in R function?

  Suppose rnorm(.) does not exist - how would you sample / simulate data from a normal distribution? (w/ $\mu=0$ and $\sigma=1$)

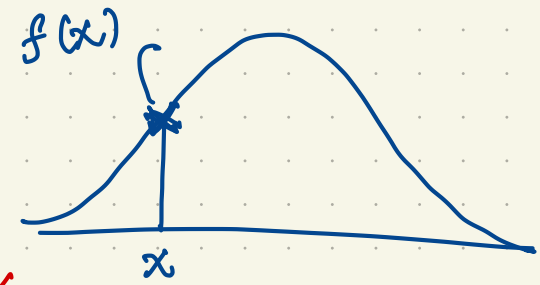
  $X \sim N(0,1)$

1. inverse transform sampling.
   - need inverse cdf to implement

"pdf" $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$.

$\uparrow$

dnorm(x, 0, 1)



$F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx$

= cannot write out a formula

2. Rejection sampling

$$\rightarrow f(x) = \cancel{\frac{1}{\sqrt{2\pi\sigma^2}}} \exp\left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$

$$\tilde{f}(x) \propto f(x)$$

no. constant
(unnormalized target)

$g(x)$: proposal density $\rightarrow$ easier distribution that you know how to sample from

$$\tilde{f}(x) \leq M\, g(x)$$

1. $U \sim \text{unif}(0,1)$
2. $\rightarrow$ $X \sim g(x)$
3. accept if $U \leq \dfrac{\tilde{f}(x)}{M g(x)}$

repeat until you have n draws

3. Importance sampling

↳ does not directly give you samples from your target

↳ estimate an expectation/probability from your target distribution

$(X \sim N(0,1))$

$P(X > 4) \longrightarrow$ rare event

$E(X^2)$

- calculate an importance weight:
- $W(x) = \dfrac{f(x)}{g(x)}$
- no loops, never throw away any samples $\longrightarrow$ computationaley efficient

1. $X \sim g$

2. $W(x) = \dfrac{f(x)}{g(x)}$

3. $\text{mean}\left(I(X > 4) \cdot w \right) \rightarrow$ estimate of $P(X > 4)$

4. SIR

sampling importance resampling

- re-use weights from importance sampling

1. M draws $\sim g$ $\rightarrow$
2. $w(x) = \dfrac{f(x)}{g(x)}$ $\quad M >> n.$

   ↳ weights.

   $M = 1000$

3. Normalize $\tilde{w}(x) = \dfrac{w(x)}{\sum_{i=1}^{M} w(x)}$
4. draw n samples according to these weights

   ↳ $\underbrace{\tilde{w}(x_1), \ldots}$ $\tilde{w}(x_M)$

   $n = 100$

sample(x, size = n, replace = TRUE, prob = $\tilde{w}$).